MIL-T-22913C(OS)
6 April 1971
Superseding
MIL-T-22913B(Wep)
22 January 1964

MILITARY SPECIFICATION

TIMER

(For Torpedo Mark 37 Mod 2)

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 Scope. This specification establishes the requirements for the procurement of a timer, NAVORD Outline Drawing 2209071.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-Z-325	Zinc Plating (electrodeposited)

Military

MIL-P-116	Preservation, Methods of
MIL-M-3171	Magnesium Alloy, Processes for corrosion Protection for
MIL-P-8585	Primer Coating; Zinc Chromate, Low Moisture sensitivity
MIL-A-8625	Anodized Coatings for Aluminum and Aluminum Alloys
MIL-W-16878	Wire; Electrical (Insulated, High Temperature)

FSC 1355

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MIL-E-82590	Environmental Requirements and Tests (For Torpedo Mk 37 Components and Assemblies)

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads, 40″ x 48″ Pallets, Skids, Runners, or Pallet-Type Base

DRAWINGS

Naval Ordnance Systems Command

NAVORD LD 162545	Control Unit (Assembled) and all parts and subassemblies listed thereon
NAVORD LD 496023	Shipping (Packing) Condition (for one timer)
NAVORD Drawing 812349	Relay (modified commercial part)
NAVORD Drawing 1563738	Relay Armature
NAVORD Drawing 696787	Control Unit assembled
NAVORD Drawing 2209071	Timer

PUBLICATIONS

Naval Ordnance Systems Command

OS 6341	Miscellaneous General Ordnance Design Requirements

(Copies of specifications, standards, drawings, and publications requested by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of the specification to the extent specified herein. Unless otherwise indicated, the

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issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards Publications

Handbook H28	Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

Consolidated Classification Committee Publication

Consolidated Freight Clarification Rules

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago, Illinois 60606.)

American Trucking Association, Inc. Publication

Motor Freight Clarification Rules

(Application for copies should be addressed to the Issuing Officer, American Trucking Association, Inc., 1424 16th St., N. W., Washington, D. C. 20036.)

3. REQUIREMENTS

3.1 Description. The timer, mounted in the control unit, NAVORD LD 162545, consists of a 400 Hertz (Hz), 115 volt root mean square (rms), three phase motor which drives, through appropriate gearing, six sets of cam-operated switches and two continuous type potentiometers.

(a) It controls the sequence and duration of application of 24 volts dc to certain torpedo control circuits on a repeat-cycle basis.

(b) It amplitude-modulates a 90 volt rms, 400 Hz signal.

(c) It amplitude-modulates a l4 volt rms, 400 Hz signal.

All external electrical connections to the timer are made through a single connector mounted on the timer frame.

3.2 General requirements

3.2.1 Data required. No data is required by this specification (other than reports accompanying samples submitted for preproduction and periodic

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production testing), or by applicable documents referenced In Section 2, unless specified in the contract or order (see 6.2).

3.2.2 Preproduction sample. Unless otherwise specified in the contract or order, preproduction samples of the timer shall be manufactured using the methods and procedures proposed for the production lot. These units shall be tested as specified in Section 4 herein and are for the purpose of determining, prior to starting production, that the contractor's production methods are capable of producing timers that comply with the technical requirements of the contract. See 403.1.

3.2.3 Materials. Unless otherwise specified In the contract, requisition, or order, all materials and purchased parts entering into the final assembly of the timer shall conform to the detailed requirements indicated on NAVORD Drawing 2209071 and to this specification.

3.2.3.1 Hookup wire. Unless otherwise specified, all hookup wire used in the timer shall conform to MIL-W-16878.

3.2.3.2 Mercury and silicones. Materials capable of giving off either mercury or silicone vapor, when subjected to temperatures within the operating range of the timer, shall not be used.

3.2.4 Finishes, coatings, construction, and dimensions. Finishes, coatings, construction, and dimensions for the timer shall be in accordance with the applicable drawings, specifications, and publications listed in NAVORD Drawing 2209071.

3.2.4.1 Aluminum alloy parts. Aluminum and aluminum alloys, if used, shall be anodized in accordance with MIL-A-8625, Type I.

3.2.4.2 Magnesium alloy parts. Magnesium and magnesium alloys, if used, shall be treated in accordance with MIL-M-3171, Type III, followed by two coats of priming per MIL-P-8585.

3.2.5 Moisture, fungus, and corrosion. The timer shall be rendered moisture, fungus, and corrosion resistant as required by OS 6341.

3.2.5.1 Ferrous metal parts. Electrical sheet steel assemblies shall be processed by accepted commercial methods to provide protection against corrosion. Where other noncorrosion resistant ferrous metals must be used, the exposed surfaces shall be zinc plated in accordance with QQ-Z-325, Type II, Class 3.

3.2.6 Marking. Marking of assemblies, subassemblies, and parts shall be in accordance with the applicable drawings listed in NAVORD Drawing 2209071.

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3.2.7 Threads. Unless otherwise specified, all threads shall be in accordance with the National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings.

3.2.8 Conflicting requirements. In the event of conflict between the documents referenced herein and the contents of this specification, the contents of this specification shall apply.

3.3 Performance requirements and product characteristics. The timer shall meet the following performance requirements and product characteristics:

3.3.1 Performance requirements

3.3.1.1 Motor input

(a) Voltage. The input voltage shall be 3 phase, 115 volts rms plus or minus 3 percent average line to line voltage, with the difference between any two line voltages not exceeding 12 volts rms.

(b) Frequency. The input frequency shall be 400 Hz plus or minus 2 percent.

(c) Power. The input power to the timer motor, when operating the steady state mechanical load imposed by the switches and potentiometers specified herein, shall not exceed 20 watts.

3.3.1.2 Starting response time. The timer motor shall have sufficient starting torque to ensure that, from every possible rest position of the cam shaft, the timer will reach its steady state operating speed within five seconds after the following applicable voltage is applied:

Three phase, 90 volts rms average line-to-line voltage maximum 400 Hz plus or minus 2 percent. The difference between any two line-to-line voltages shall not exceed 12 volts. Steady state operation shall be considered to exist if circuit 1 repeat-cycles in 1250 milliseconds (ma) plus or minus 27 ms.

3.3.1.3 Switch circuits

3.3.1.3.1 Circuit time. The timer motor, energized as specified in 3.3.1.1(a) and 3.3.1.1(b), shall cause one cycle of operation in each circuit listed in Table I to be 1250 plus or minus 27 ms and repeatable within 2 ms. The time required to complete one cycle of operation in circuits 6 and 7, listed in Table II, shall be 20 seconds plus or minus 220 ma and repeatable within 32 ms.

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3.3.1.3.2 Switching functions. With the timer motor operating on an input frequency of 400 Hz, and on the applicable input voltage specified in 3.3.1.1(a), the timer shall perform the switching functions specified in Tables I and II. Unless otherwise specified, all times listed in Table I are with respect to the closure of circuit 1. For corrections in time due to variations in frequency, see 4.5.4.3.2.

3.3.1.3.3 Contact ratings. Switches shall be capable of making and breaking the load current specified in Table I for the circuit in which they are used at a voltage of 26 volts dc plus or minus 10 percent and in Table II for the circuit in which they are used at a voltage of 7 volts plus or minus 10 percent, 400 Hz plus or minus 10 percent, and or 3 volts dc plus or minus 10 percent.

3.3.1.3.4 Switch circuit voltage drop. Switch circuit voltage drop shall not exceed:

(a) Two volts dc at the current specified in Table I when measured at the connector terminals.

(b) One-half volt, 400 Hz at the current specified in Table II when measured at the connector terminals.

3.3.1.3.5 Contact bounce. The duration of contact bounce occurring in any circuit at switch closure shall not exceed one ms for circuits 1 through 5 inclusive and three ms for circuits 6 and 7. For purposes of time measurement, a circuit shall be considered closed when the switch contacts meet for the first time during each repeat-cycle.

3.3.1.4 Dielectric requirements

3.3.1.4.1 DC and potentiometer 155 circuits. The switches and the 0 to 14 volt potentiometer 155 shall be capable of withstanding 500 volts rms, 60 Hz between terminals P152-9 through P152-22 (inclusive) and the timer chassis, with the motor and 90 volt potentiometer terminals, P152-1 through P152-7 (inclusive), grounded to the chassis. The test voltage shall start at zero volts, be raised to 500 volts rms within 20 seconds, and then maintained for one minute. No arcing or breakdown of insulation shall occur.

TABLE I - SWITCHING DATA

Circuit	Closes (ms)	Opens (ms)	Maximum Steady State Relay Coil Current (Amperes dc)
1	0	52 ±2	0.8

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TABLE I - SWITCHING DATA (Cont'd)

Circuit	Closes (ms)	Opens (ms)	Maximum Steady State Relay Coil Current (Amperes dc)
2	625 ±6	52 ±2 after circuit 2 closes	0.8
3	10 ±3 after circuit 1 opens	52 ±2 after circuit 3 closes	0.1
4	30 ±10	445 ±10	0.1
5	0 (Arbitrary)	625 ±15	0.5

TABLE II - SWITCHING DATA

Circuit	Closes (ms)	Opens (ms)	Maximum Steady State 400 Hz current (Amperes rms)
6	0 + 28 after circuit 7 opens	10,000 ±110	0.030
7	0 + 28 after circuit 6 opens	10,000 ±110 after circuit 7 closes	0.030

3.3.1.4.2 AC circuits. The motor and the 0 to 90 volt potentiometer 154 shall be capable of withstanding 1000 volts rms, 60 Hz between their terminals, P152-1 through P152-7 (inclusive), and the timer chassis, with all 0 to 14 volt potentiometer and switch terminals, P152-9 through P152-22 (inclusive), grounded to the chassis. The test voltage shall start at zero volts, be raised to 1000 volts rms within 20 seconds, and then maintained for one minute. No arcing or breakdown of insulation shall occur.

3.3.1.4.3 Insulation resistance. Insulation resistance at 500 volts dc minimum shall. be 50 megohms minimum between all terminals of the connector and the timer chassis and all terminals of the connector and its shell.

3.3.1.5 Operating time. The timer shall be capable of operating for 45 minutes plus or minus 5 minutes, followed by sufficient time to cool to ambient temperature.

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3.3.1.6 Operating life. The timer shall have a useful life of at least 200 hours without lubrication or overhaul.

3.3.2 Product characteristics

3.3.2.1 Adjustment. The timer shall be designed so that the sequence of circuit switching can be adjusted through 360 degrees. The length of switch closure time for each circuit can be adjusted as defined in Table III.

TABLE III - PULSE LENGTH ADJUSTMENT TABLE

Circuit	Adjustment
1	25-150 ms
2	25-150 ms
3	25-150 ms
4	415 ±150 ms
5	625 ±100 ms
*6	45% to 55% of time specified in one cycle of potentiometer operation
*7	55% to 45% of time specified in one cycle of potentiometer operation

*The sum of the pulse length adjustment for circuits 6 and 7 shall equal. the time for one cycle of potentiometer operation.

3.3.2.2 90-volt potentiometer, potentiometer 154. Potentiometer 154 shall be capable of continuous rotation and shall have the following characteristics:

(a) Wattage rating. Three watts minimum, continuous at 154 degrees centigrade.

(b) Continuity. Wiper shall maintain contact with winding through 360 degrees of rotation.

(c) Tap points. The potentiometer shall have four tap points on its winding as specified in the following: Resistance between tap points 5000 ohms plus or minus 5 percent.

(d) Tap position. Zero degrees; 90 degrees plus or minus 2 degrees; 180 degrees plus or minus 2 degrees; 270 degrees plus or minus 2 degrees. The end of the winding shall be connected to the start of the winding at the zero degrees tap. The end of the winding shall be within two mechanical degrees of the start of the winding.

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(e) Linearity. Plus or minus one percent.

(f) Resolution. A minimum of 1000 turns of wire per potentiometer.

(g} The potentiometer shall be capable of withstanding 1000 volts rms, 60 Hertz between its terminals and its chassis mount.

(h) The potentiometer shall be capable of withstanding any environmental condition that may result due to testing the timer as specified in 3.4.

3.3.2.3 14-volt potentiometer, potentiometer 155. Potentiometer 155 shall be capable of continuous rotation and shall have the following characteristics:

(a) Wattage rating. One watt minimum, continuous at 54 degrees centigrade.

(b) Continuity. Wiper shall maintain contact with winding through 360 degrees of rotation.

(c) Tap points. The potentiometer shall have four tap points on its winding as specified in the following: Resistance between tap points 1000 ohms plus or minus 5 percent.

(d) Tap position. Zero degrees; 90 degrees plus or minus 2 degrees; 180 degrees plus or minus 2 degrees; 270 degrees plus or minus 2 degrees. The end of the winding shall be connected to the start of the winding at the zero degrees tap. The end of the winding shall be within two mechanical degrees of the start of the winding.

(e) Linearity. Plus or minus one percent.

(f) Resolution. A minimum of 1000 turns of wire per potentiometer.

(g) The potentiometer shall be capable of withstanding 500 volts rms, 60 Hertz between its terminals and its chassis mount.

(h) The potentiometer shall be capable of withstanding any environmental condition that may result due to testing the timer as specified in 3.4.

3.3.2.4 Internal conditions. Switch circuits 6 and 7 shall be connected to the 0 degree and 180 degrees tap of potentiometer 155 as shown in NAVORD Drawing 2209071. There shall be no other electrical connections made between the motor circuit, any of the switch circuits, and either of the potentiometer circuits. No circuit within the timer shall be connected to the timer frame.

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3.3.3 Mechanical alignment

3.3.3.1 Potentiometer wiper arms. With the two potentiometers geared together, the wiper arms shall be no more than 2.5 degrees out of phase and will not change when subjected to the conditions specified In 3.4.

3.3.3.2 Switch circuits 6 and 7. Switch circuit 6 shall break and switch circuit 7 shall make within those limits specified in Table II, one plus 0.05, minus 0.25 second before the wiper arm in potentiometer 155 reaches 90 degrees, as defined in 4.5.4.3(e), when the repeat cycle time of circuits 6 and 7 meets the requirements of 3.3.1.3.1. Switch circuit 7 shall break and switch circuit 6 shall make within those limits specified in Table II, one plus 0.05, minus 0.25 second before the wiper arm in potentiometer 155 reaches 270 degrees, as defined in 4.5.4.3(c), when the repeat cycle time of circuits 6 and 7 meets the requirements of 3.3.1.3.1. There shall be a break-before-make condition in the switching of circuits 6 and 7.

3.4 Environmental requirements. The timer shall meet the other applicable requirements of this specification during or after the tests defined or specified in 4.5.

3.4.1 Vibration. The timer shall meet the following vibration requirements:

3.4.1.1 Vibration, nonoperating. The timer shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.4.1.2.

3.4.1.2 Vibration, operating. The timer shall show no damage and shall meet all the performance requirements and product characteristics of this specification during and after the test specified in 4.4.1.3.

3.4.2 Temperature and humidity

3.4.2.1 High temperature and humidity, nonoperating. The timer shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.4.2.1.

3.4.2.2 High temperature, operating. The timer shall be capable of meting all the performance requirements of this specification while undergoing the test specified in 4.4.2.2.

3.4.2.3 Low temperature, nonoperating. The timer shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.4.2.3.

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3.4.2.4 Low temperature, operating. The timer shall be capable of meeting all the performance requirements of this specification while undergoing the test specified in 4.4.2.4.

305 Cleanup. Prior to and after final assembly, all parts, components, and the assembly shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign matter. Burrs and sharp edges, as well as rosin flash, shall be removed.

3.6 Workmanship. The equipment, including all parts and accessories, shall be constructed and finished in a reamer to ensure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, marking of parts, and assemblies, impregnation of coils, plating, painting, machine screw assemblage, welding, riveting, brazing, and freedom of parts from all burrs and sharp edges. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspection of the timer shall be classified as follows:

(a) Reproduction inspection (paragraph 4.4)

(b) Quality conformance inspection (paragraph 4.5)

(c) Periodic production inspection (paragraph 4.4)

4.3 Sampling

4.3.1 Preproduction. A preproduction sample of five timers, manufactured in accordance with 3.2.2, shall be subjected to the preproduction tests detailed in 4.4 at an activity designated by the procuring activity.

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Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the timer by the supplier, prior to the approval of the preproduction sample, shall be at the supplier's risk.

4.3.2 Lot size. Unless otherwise specified in the contract or order, each shipment of units of product shall be a separate "inspection lot" and may differ from the quantity specified in the total order or contract. In any event, all units of a single "inspection lot" shall be made to the same design, with the same materials and manufacturing techniques, and shall be part of one continuous production run.

4.3.3 Periodic production samples. From each succeeding lot or grand lot of 300 or less that have passed the quality conformance inspection of 4.5, a random sample of five timers will be selected for submission to an activity designated by the procuring activity to determine compliance of the samples with the requirements of the contract, specification, and drawings. The timers shall be considered satisfactory if they meet all specification and pertinent drawings requirements and withstand the tests specified in 4.4 and 4.5. Failur of the periodic production sample to comply with these requirements will result in the rejection of the lot, as determined by the procuring activity. Periodic production samples accepted will be applied as part of the quantity specified by the contract or order, but shall not be used in any equipment manufactured for the Government.

4.4 Preproduction and periodic production inspection. The preproduction and periodic production samples, after satisfactorily passing the quality conformance inspections detailed In 4.5, shall be subjected to the following tests and examinations. Certified inspection data covering the results of preproduction and periodic production inspections shall be forwarded to the procuring activity.

4.4.1 Vibration

4.4.1.1 Mounting. The timer, less pack or packaging, shall be mounted by means of its mounting holes as shown on NAVORD Dwg 2209071 and tested as specified in 4.4.1.2 and 4.4.1.3.

4.4.1.2 Vibration, nonoperating. The timer shall be subjected to the "Vibration, nonoperating" test of MIL-E-82590 and thereafter shall meet the requirements of 3.4.1.1.

4.4.1.3 Vibration, operating. The timer shall be subjected to the "Vibration, operating" test of MIL-E-82590 and during and thereafter shall meet the requirements of 3.4.1.2.

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4.4.2 Temperature and humidity

4.4.2.1 High temperature and humidity, nonoperating. The timer shall be subjected to the "High temperature and humidity, nonoperating" test of MIL-E-82590 and thereafter shall meet the requirements of 3.4.2.1.

4.4.2.2 High temerature, operating. The timer shall be subjected to the "High temperature, operating" test of MIL-E-82590 and, while at the test temperature, shall meet the requirements of 3.4.2.2.

4.4.2.3 Low temperature, nonoperating. The timer shall be subjected to the "Low temperature, nonoperating" test of MIL-E-82590 and thereafter shall meet the requirements of 3.4.2.3.

4.4.2.4 Low temperature, operating. The timer shall be subjected to the "Low temperature, operating" test of MIL-E-82590 and, while at the test temperature, shall meet the requirements of 3.4.2.4.

4.4.3 Motor input power. Apply the applicable input voltage, as specified in 3.3.1.1 (a), to the timer motor. The power drawn by the timer motor when driving both the switch and the potentiometer loads shall not exceed the limit specified in 3.3.1.1.

4.4.4 Starting response time. Connect the coil of a relay per NAVORD Drawing 812349 in series with timer circuit 1. Connect the remaining circuit 1 timer terminal and the remaining terminal on the relay coil across a 26 volt dc plus or minus 10 percent power supply. Record the operation of circuit 1 from the time the applicable voltage and frequency specified in 3.3.1.2 is applied to the timer motor until steady state operation occurs in circuit 1. Repeat this test 24 times. Advance the cam shaft rest position approximately 15 degrees, for each successive test, from the rest position it had at the start of the preceding test. The timer shall perform as specified in 3.3.1.2.

4.4.5 Switch contact tests

4.4.5.1 Contact voltage drop. Connect the coil of a relay per NAVORD Drawing 812349 to the common terminal of circuits 1 and 2. Connect the coil of a 12-2-13001-9431 relay in series with circuit 4; connect the coil of a Relay, NAVORD Dwg 1563738, in series with circuit 3. Join the remaining terminals on all relay coils together and connect them to the zero volt dc terminal of a 26 volt dc plus or minus 10 percent power supply. For stitch circuit 5, connect terminal 16 on P-152 to the plus 26 volt terminal of the dc power supply, connect terminal 17 on P-152 to a 48 ohm resistor, and connect the other side of the resistor to the zero volt dc terminal of the dc power supply. Join the remaining stitch circuit terminals of circuits 1 through 4 together and connect them to the plus 26 volt dc terminal of the power supply. Apply the applicable voltage and frequency specified in 3.3.1.1 to the timer motor. For switch circuits 6 and 7 connect terminals 18 and 19 of P152 and one side of a

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230 ohm plus or minus 10 percent resistor together. Connect the other side of the resistor to an output terminal of the 5-10 volt 400 Hz single phase power supply. Connect the other terminal of the power supply to either 20 or 21 of P152. Adjust the power supply to 7 volts rms plus or minus 10 percent. Measure the voltage drop across each switch in the timer while the switch is carrying load current. The total drop in each timer circuit while the circuit is closed shall not exceed the limits specified in 3.3.1.3.4.

4.4.5.2 Contact bounce. Using the identical test setup specified in 4.4.5.1, measure the duration of contact bounce which may occur in the closure of each circuit. Contact bounce may be observed and measured by placing the input terminals of an oscilloscope, capable of driven-sweep operation, across either the switch being tested or the load impedance connected in series with the switch. A standard one kHz per second oscillator may be used to calibrate the oscilloscope. The duration of the contact bounce in any timer circuit shall not exceed the limit specified in 3.3.1.3.5.

4.4.6 Potentiometer test. The potentiometers shall be checked for compliance with 3.3.2.2(d), 3.3.2.2(e), 3.3.2.2(f), 3.3.2.3(d), 3.3.2.3(e), and 3.3.2.3(f).

4.4.7 Operating life. For this test, a new timer may be used or a timer that has been under test and overhauled may be used. Use the test setup specified in 4.4.5.1 and, in addition, connect a source of 90 volts plus or minus 3 percent, 400 Hz plus or minus 2 percent to the voltage supply terminals of the 90 volt potentiometer, and a source of 14 volts plus or minus 3 percent, 400 Hz plus or minus 2 percent to the voltage supply terminals of the 14 volt potentiometer. Operate the timer for a total of 199 hours, always maintaining the input voltages and frequency within the specified limits. At the end of each 45 minutes plus or minus 5 minutes of operation, stop the timer. If the manufacturer so requests, the timer shall be allowed to cool to ambient temperature before it is again started and operated as described above. Once during each 10 hours of operation, the timer operation shall be checked as specified in 4.5. During this test, no maintenance shall be performed on the timer. While the timer is operating in the steady state condition, performance shall be within limits specified in 3.3.1.1(c), 3.3.1.3, and 3.3.3.1. When the timer is started, performance shall be within limits specified in 3.3.1.2.

4.5 Quality conformance inspection. Each timer delivered under contract shall be subjected to the following tests and examinations. Failure to meet any requirement specified herein shall be considered cause for rejection.

4.5.1 Test conditions. Unless otherwise specified, the timer shall be subjected to quality conformance tests under the following conditions:

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4.5.1.1 Temperature: Room ambient 65 degrees F to 95 degrees F (18.33 degrees C to 35 degrees C).

4.5.1.2 Altitude: Normal ground.

4.5.1.3 Vibration: Normal.

4.5.1.4 Humidity: From O up to 95 percent relative maximum, inclusive.

4.5.2 Test and inspection equipment and facilities. The manufacturer shall furnish and maintain all necessary test equipment and facilities and shall provide personnel far performing all quality conformance tests. The test equipment shall be adequate In quantity and, when definite requirements are not specified shall be of sufficient accuracy and quality to permit performance of the required tests. The following test equipment is required to perform the quality conformance tests of this specification:

4.5.2.1 AC power supply. (a) Three phase 85 to 120 volts rms adjustable, 400 Hz plus or minus 2 percent, 0.5 ampere per line maximum required. (b) Single phase 5 to 10 volts rms minimum range, 400 Hz plus or minus 2 percent, 0.1 ampere rms maximum required.

4.5.2.2 DC power supply. (a) 26 volts plus of minus 10 percent, 2.0 amperes maximum required. (b) Six volts plus or minus 10 percent, 15 milliamperes maximum required. This power supply must also be capable of supplying plus or minus 3 volts dc relative to ground.

4.5.2.3 AC voltmeter, 0-150 volts rms at 400 Hz, minimum impedance 3000 ohms.

4.5.2.4 Dc voltmeter, 0-30 volts, minimum resistance 1000 ohms per volt.

4.5.2.5 Wheat stone bridge, 0-9999 ohms minimum, in steps of one ohm maximum.

4.5.2.6 Electronic counter

4.5.2.6.1 Frequency measurement section.

4.5.2.6.1.1 Range: 390 to 410 Hz minimum.

4.5.2.6.1.2 Accuracy: Plus or minus 0.1 Hz minimum.

4.5.2.6.1.3 Allowable input signal voltage: 1.20 volts rms. This input signal voltage may be reduced, by means of a potentiometer, to a

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level suitable for any otherwise acceptable commercial counter. If a potentiometer is used, its total resistance shall not be less than the impedance specified in 4.5.2.6.1.4.

4.5.2.6.1.4 Input impedance at 400 Hz: 5000 ohms minimum.

4.5.2.6.2 Time interval measurement section.

4.5.2.6.2.1 Range: 0.1 ms to 1500 ms minimum.

4.5.2.6.2.2 Accuracy: Plus or minus 0.1 ms minimum.

4.5.2.6.2.3 Triggering circuits. The start-to-count signal channel and the stop-count signal channel shall be capable of either independent or common operation. The change from independent to common operation shall be made by means of a switch.

(a) Signal range: Both the start and stop channels shall be capable of operating on signals having a positive slope and a magnitude of plus or minus 30 volts dc. The triggering level shall be continuously adjustable from plus 30 volts dc to minus 30 volts dc minimum.

(b) Input impedance: 30,000 ohms shunted by 100 micromicrofarads minimum.

4.5.2.6.3 Period measurement section.

4.5.2.6.3.1 Range: 18 seconds to 22 seconds minimum.

4.5.2.6.3.2 Accuracy: plus or minus 0.01 seconds minimum.

4.5.2.6.3.3 Allowable input signal voltage: 15 volts peak.

4.5.2.6.3.4 Input impedance: 50,000 ohms shunted by 100 micromicrofarads.

4.5.2.7 Four dummy loads, constricted as follows:

(a) One resistive leg shall be connected in parallel with a capacitive-resistive leg. The junction between the two resistors shall be called terminal A.

(b) The remaining connection between the parallel legs shall be called terminal B.

(c) The connections between the resistor and capacitor in the capacitive-resistive leg shall be called terminal C.

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4.5.2.7.1 Resistive leg: 330 ohms plus or minus 10 percent 2 watt resistor.

4.5.2.7.2 Capacitive-resistive leg: 2 microfards plus 20 percent minus 10 percent, paper dielectric, 50 volt dc minimum, capacitor; connected in series with a 3300 ohms plus or minus 10 percent, 1/2 watt resistor.

4.5.2.8 High potential test set: This set shall be capable of furnishing 500 volts rms and 1000 volts rms at 60 Hz. The set will be used for insulation testing and shall include an alarm bell or light or combination of both to indicate when any arcing or breakdown occurs.

4.5.2.9 Recorder: Sanborn Model 320, dual channel dc amplifier recorder or equivalent.

4.5.2.10 Switch: one three pole, single throw switch is required. The switch shall have sufficient insulation resistance and creepage distance between adjacent poles to withstand 1000 volts rms at 60 Hz for one minute without any arcing or breakdown of insulation. Each switch pole shall be capable of carrying a one-half ampere minimum reactive load current at 120 volts rms, 400 Hz continuously without any evidence of overheating. Maximum contact drop at one-half ampere rms shall not exceed 0.1 volt rms.

4.5.2.11 Oscilloscope: Tektronix Type 515 or equivalent.

4.5.2.12 Two resistors: 5100 ohms each, plus or minus 5 percent, 1/2 watt carbon.

4.5.2.13 Insulation tester capable of measuring 50 megohms with 500 volts dc applied.

4.5.3 Test setup

4.5.3.1 Circuit setting adjustments shall be locked in place.

4.5.3.2 Using a separate dummy load for each connection, connect the B terminal of a dummy load to each of the following: P-152 terminal 11, P-152 terminal 9, a terminal of circuit 3, and a terminal of circuit 4.

4.5.3.3 Connect the A terminals of all dummy loads to the negative terminal of the plus 26 volt dc power supply.

4.5.3.4 Join terminal 10 on P-152 and the remaining terminals of circuits 3 and 4 together, and then connect to the plus 26 volt dc terminal of the dc power supply.

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4.5.3.5 Connect terminal 17 of P-152 to a 330 ohm plus or minus 10 percent resistor. Connect the other end of the resistor to the plus 26 volt terminal of the dc power supply. Connect terminal 16 of P-152 to the negative terminal of the dc power supply.

4.5.3.6 Connect terminal 20 of P152 to center-tapped common ground of 6 volt dc power supply. Connect terminal 18 of P152 to positive terminal of the power supply, and connect terminal 19 of P152 to the negative terminal.

4.5.3.7 Connect the timer motor input terminals to the load side of the three pole, single throw switch. The switch shall be In the open position. Connect the input side of the switch to the 85-120 volt ac power supply.

4.5.4 Test procedure. See NAVORD Drawing 2209071 for timer terminal. designations.

4.5.4.1 Voltage measurements

4.5.4.1.1 Motor input voltage. Using the ac voltmeter, set the ac power supply voltage as specified in 3.3.1.2. Close the three pole, single throw switch. The timer shall start functioning without any additional external aid. Open switch and repeat the above procedure a total of four times, all at different initial cam positions. Reset the motor input voltage to the value specified in 3.3.1.1.

4.5.4.1.2 Switch circuit voltage. Using the dc voltmeter, measure and record the voltage supplied to the timer switch circuits. It shall be as specified in 3.3.1.3.3.

4.5.4.2 Frequency. Using the frequency measuring section of the electronic counter, measure and record the timer motor input frequency. It shall be as specified in 3.3.1.1.

4.5.4.3 Switch circuit time measurements

(a) For tests 1 through 8. Connect the ground terminal of the time interval section of the electronic counter to the zero volt terminal of the dc power supply. Connect the “start of count” and “stop counting” leads of the electronic counter to terminal C only of the applicable dummy load for each test specified in Table IV. Where positive triggering voltage polarities are specified in Table IV, the electronic counter trigger slope setting shall be positive. Where negative triggering voltage polarities are specified in Table IV, the electronic counter trigger slope settings shall be negative. Triggering levels on the electronic counter shall be three volts plus or minus one volt dc less than the voltage measured in

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4.5.4.1.2. Measure and record the maximum and minimum time intervals for at least four cycles, as read by the electronic counter, for each test specified In Table IV. In addition to meeting the time limit specified for test 8, the reading shall be repeatable within 2 ms.

(b) For test 9 (circuit 5). Connect terminal 17 of P-152 to the "start of count" and stop of count" leads of the electronic counter. Connect the ground terminal of the timer interval section of the electronic counter to the zero volt terminal of the dc power supply. Triggering level on the electronic counter shall be positive three volts plus or minus one volt dc less than the voltage measured in 4.5.4.1.2. Measure and record the maximum and minimum time intervals for at least four cycles, as read by the electronic counter. All other timing circuits are disconnected from the counter during test 9.

TABLE IV. ELECTRONIC COUNTER SETTINGS

	"Start of count" channel	"Stop counting"	Channel
Test	Connect to Trigger Polarity	Connect to Trigger Polarity	Operation
1	circuit 1 positive	- - - - - negative	common
2	circuit 1 positive	circuit 2 positive	independent
3	circuit 2 positive	- - - - - negative	common
4	circuit 1 negative	circuit 3 positive	independent
5	circuit 3 positive	- - - - - negative	common
6	circuit 1 positive	circuit 4 positive	independent
7	circuit 1 positive	circuit 4 negative	independent
8	circuit 1 positive	- - - - - positive	see 4.5.4.3.1
9	see 4.5.4.3(b)		

(c) For tests 11 and 13. Connect the negative terminal of the 6 volt power supply to 19-P152 and the positive terminal to 18-P152. Join the ground terminal of the oscilloscope to the center tap of the power supply and connect to 20-P152. Connect 21-P152 to the input terminal of the oscilloscope. Set the sweep at 20 ms per centimeter; adjust sweep, if necessary, to obtain best reading. For test 11, set the oscilloscope to trigger internally on a negative slope, dc signal; for test 13, trigger

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on a positive slope, dc signal. Because the wave form will be of short duration in both cases, observe it for several cycles in order to obtain an accurate reading.

(d) For tests 10 and 12. Connect the negative terminal of the 6 volt dc power supply to 21-P152 and the positive terminal of 20-P152. Connect 22-P152 to the ground terminal. of the dual channel paper strip recorder, and connect 19-P152 to channel 1. Connect 18-P152 to channel 2. Connect one 5100 ohm resistor between 18-P152 and ground. Connect one 5100 ohm resistor between 19-P152 and ground. Set the recorder to a chart speed of 100 mm/second, 0.2 volt/mm, turn on the timer marker and record at least one complete cycle of operation. Test 10 results may be read on channel 1 and test 12 results on channel 2.

(e) Using the recordings taken in section (d) of this paragraph, measure the switching time of circuits 6 and 7 relative to the motion of potentiometer 155. On channel 1 measure the time between the switching "on" of circuit 7 to the maximum positive excursion, and on channel 2 the time between the switching "on" of circuit 6 to the maximum negative excursion. These measurements should be as specified in 3.3.3.2.

4.5.4.3.1 Set channel operation to "independent." Reset the electronic counter so that zero time is read. Within 0.5 second maximum after the electronic counter begins to count due to the application of the start-to-count triggering voltage, switch channel operation to "common."

4.5.4.3.2 Test data correction. The data taken as specified in Table IV shall be corrected for timer motor input frequency as follows:

$$\text{Correct reading} = \frac{(\text{actual reading})\ (\text{frequency of 4.5.4.2})}{400\ \text{Hz}}$$

4.5.4.3.3 Test results. The frequency corrected results of 4.5.4.3.2 shall be as follows. The data listed below is taken from Tables I and II and 3.3.1.3.1, and repeated here for convenience:

Test 1	52 ±2 ms	Test 7	445 ±10 ms
Test 2	625 ±6 ms	*Test 8	1250 ±27 ms
Test 3	52 ±2 ms	Test 9	625 ±15 ms
Test 4	10 ±3 ms	Test 10	10,000 ±110 ms
Test 5	52 ±2 ms	Test 11	0 + 28 ms, - 0 ms
Test 6	30 ±10 ms		
		Test 12	10,000 ±110 ms

Test 13 0 + 28 ms, - 0

* See repeat cycle reference at end of 4.5.4.3(a)

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4.5.4.4 Potentiometer operation

4.5.4.4.1 Test setup. Join 4-P152 and 20-P152 and connect to positive terminal of 6-volt dc power supply. Join 5-P152 and 21-P152 together and connect to negative side of 6-volt dc power supply. Join 18-P152, 19-P152, and 6-P152 together and connect to ground terminal of the dual channel paper strip recorder. Connect 90-volt potentiometer (7-P152) to channel 1 and connect 14-volt potentiometer (22-P152) to channel 2. With the recorder chart speed set to 20 mm/sec and the gain set to 0.2 volt/mm, record at least two complete cycles of timer operation.

4.5.4.4.2 90-volt potentiometer operating. Using the recording made in 4.5.4.4.1, observe the rise and fall of the potentiometer output voltage. The oscillograph recording of the voltage variation shall be linear with no discontinuities in the trace.

4.5.4.4.3 14-volt potentiometer operation. (a) Using the recording made in 4.5.4.4.1, observe the rise and fall of the potentiometer output voltage. The oscillograph recording of the voltage variation shall be linear with no discontinuities in the trace. (b) The phase measurement shall be made by comparison between channels 1 and 2 of the recording made in 4.5.4.4.1, and shall be as specified in 3..3.3.1.

4.5.4.5 Potentiometer resistance

4.5.4.5.1 90-volt potentiometer resistance. Using the Wheatstone bridge, measure the potentiometer resistance between terminals 4 and 6 and 5 and 6 on P-152. Each reading shall be 2500 ohms plus or minus 5 percent.

4.5.4.5.2 14-volt potentiometer resistance. Using the Wheatstone bridge, measure the potentiometer resistance between terminals 20 on P-152 and either 18 or P152 or 19 on P-152 and 21 on P-152 and either 18 on P-152 or 19 on P-152. Each reading shall be 500 ohms plus or minus 5 percent. If measuring from 18 on P-152, the resistance is 500 ohms. The difference between the two readings shall be 5 ohms or less. From 19 on P-152, without timer rotation, the circuit will be open. If measuring from 19 on P-152, the resistance is 500 ohms; from 18 on P-152, the circuit will be open with no timer rotation.

4.5.4.6 Dielectric tests. Apply test voltages of the timer terminals, as specified in 3.3.1.4. No arcing or breakdown of insulation shall occur.

4.5.4.7 Measure the insulation resistance between all terminals of the connector and the timer chassis and all terminals of the connector and its shell at 500 volts dc minimum. The insulation resistance shall be 50 megohms minimum.

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4.5.4.8 Visual and dimensional examination. Each timer shall be examined to determine conformance with the requirements for workmanship and dimensions.

4.5.4.9 Packaging, packing, and marking. Packaging, packing, and marking shall comply with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. The levels of preservation and packaging shall be as specified (see 6.2).

5.1.1 Level A

5.1.1.1 Unit packaging. The timer shall be packaged in accordance with Method IID of MIL-P-116 and LD 496023.

5.1.2 Level B. Not applicable

5.1.3 Level C

5.1.3.1 Unit packaging. The timer shall be packaged in accordance with Method III of MIL-P-116.

5.2 Packing. Packing shall be Level A or Level C, as specified (see 6.2).

5.2.1 Number of packages per pack. The number of unit packages per pack, for Level C, shall be as specified in the contract or order.

5.2.2 Level A

5.2.2.1 Unit pack. The timer shall be packed in accordance with LD 496023 and may be shipped individually in small lots.

5.2.2.2 Palletizing. When specified, Level A packs shall be palletized in accordance with MIL-STD-147, Load Type III (See 6.2).

5.2.3 Level B. Not applicable.

5.2.4 Level C. The timer shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with common carrier regulations applicable to the mode of transportation.

5.3 Marking

5.3.1 Normal markings, Level A. Unit packs shall be marked in accordance

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with MIL-STD-129 and LD 496023.

5.3.2 Normal markings, Level C. Unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

5.3.3 Serial numbers. Unit packages and shipping containers shall be marked with the serial number of the units contained within.

6. NOTES

6.1 Intended use. The timer is intended for use in Mk 37 Mod 2.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Any exceptions to this specification.

(c) Required level of preservation and packaging (see 5.1).

(d) Required level of packing, including palletizing (see 5.2).

(e) Assigned activity for preproduction and periodic production inspections.

(f) Data required other than that specified in 3.2.1.

(g) That additional samples required because of failure of any unit(s) to pass the prescribed periodic production inspections shall be provided and inspected at the expense of the supplier.

Preparing Activity
Navy - OS
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